

# **Validation of Single-Camera MediaPipe BlazePose for Knee Joint Angle Measurement: Concurrent Validity, Inter-Rater Reliability, and Exploratory Test-Retest Reliability Across Multiple Functional Tasks**

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# Validation of Single-Camera MediaPipe BlazePose for Knee Joint Angle Measurement: Concurrent Validity, Inter-Rater Reliability, and Exploratory Test-Retest Reliability Across Multiple Functional Tasks

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## Abstract

**Background:** Markerless pose estimation offers a cost-free approach to remote knee joint assessment, but multidimensional validation across reliability dimensions is lacking.

**Objective:** To evaluate the concurrent validity, inter-rater reliability, and test-retest reliability of the MediaPipe BlazePose Heavy model for knee joint angle measurement across three functional tasks (Squat, Sit-to-Stand, and bilateral Single-Leg Stance, the latter performed on each leg).

**Methods:** Fifteen healthy adults (8 male, 7 female; age 20.8 +/- 0.6 years; BMI 23.9 +/- 3.5 kg/m<sup>2</sup>) performed Squat, Sit-to-Stand, and bilateral Single-Leg Stance tasks while filmed from a lateral view. Static knee flexion was measured at four goniometer-set angles (30 degrees, 60 degrees, 90 degrees, 120 degrees). BlazePose task-specific knee angles were compared with Kinovea 2D video annotation for dynamic tasks and with goniometry for static positions using intraclass correlation coefficients (ICC[2,1]), Bland-Altman analysis, and RMSE. Four participants returned for a retest session.

**Results:** Inter-rater Kinovea reliability was excellent (ICC = 0.999 [0.998, 0.999]). Dynamic concurrent validity was excellent overall (ICC = 0.982 [0.95, 0.99], MAE = 7.07 degrees) but with a systematic underestimation of -5.12 degrees; task-stratified ICCs were 0.773 [0.23, 0.91] (Squat) and 0.632 [-0.08, 0.87] (Sit-to-Stand). Static accuracy was angle-dependent (RMSE = 15.49 degrees), with overestimation at shallow flexion and underestimation at deep flexion. Single-leg stance showed poor validity (ICC = 0.136 [-0.15, 0.40]), with task-intrinsic instability supported by negative Kinovea-self retest ICCs. In an exploratory test-retest sub-sample (n = 4 participants), the BlazePose-vs-Kinovea bias was not stable across sessions (ICC = 0.094 [-0.22, 0.40]), suggesting that fixed-offset calibration should not be assumed and requires confirmation in a larger cohort.

**Conclusions:** BlazePose shows promising concurrent validity for dynamic flexion-focused tasks but exhibits clearly characterized failure modes-angle-dependent static bias, SLS instability, and session-to-session bias drift-that define the boundaries of its applicability in telerehabilitation. These preliminary boundaries may inform future protocol design and screening-level telerehabilitation use, pending larger-sample and clinical-population validation. Clinical Trial: Not applicable.

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## Original Manuscript

# Validation of Single-Camera MediaPipe BlazePose for Knee Joint Angle Measurement: Concurrent Validity, Inter-Rater Reliability, and Exploratory Test–Retest Reliability Across Multiple Functional Tasks

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## Abstract

**Background:** Markerless pose estimation offers a cost-free approach to remote knee joint assessment, but multidimensional validation across reliability dimensions is lacking.

**Objective:** To evaluate the concurrent validity, inter-rater reliability, and test–retest reliability of the MediaPipe BlazePose Heavy model for knee joint angle measurement across three functional tasks (Squat, Sit-to-Stand, and bilateral Single-Leg Stance, the latter performed on each leg).

**Methods:** Fifteen healthy adults (8 male, 7 female; age  $20.8 \pm 0.6$  years; BMI  $23.9 \pm 3.5$  kg/m<sup>2</sup>) performed Squat, Sit-to-Stand, and bilateral Single-Leg Stance tasks while filmed from a lateral view. Static knee flexion was measured at four goniometer-set angles (30°, 60°, 90°, 120°). BlazePose task-specific knee angles were compared with Kinovea 2D video annotation for dynamic tasks and with goniometry for static positions using intraclass correlation coefficients (ICC[2,1]), Bland–Altman analysis, and RMSE. Four participants returned for a retest session.

**Results:** Inter-rater Kinovea reliability was excellent (ICC = 0.999 [0.998, 0.999]). Dynamic concurrent validity was excellent overall (ICC = 0.982 [0.95, 0.99], MAE = 7.07°) but with a systematic underestimation of  $-5.12^\circ$ ; task-stratified ICCs were 0.773 [0.23, 0.91] (Squat) and 0.632 [−0.08, 0.87] (Sit-to-Stand). Static accuracy was angle-dependent (RMSE = 15.49°), with overestimation at shallow flexion and underestimation at deep flexion. Single-leg stance showed poor validity (ICC = 0.136 [−0.15, 0.40]), with task-intrinsic instability supported by negative Kinovea-self retest ICCs. In an exploratory test–retest sub-sample (n = 4 participants), the BlazePose-vs-Kinovea bias was not stable across sessions (ICC = 0.094 [−0.22, 0.40]), suggesting that fixed-offset calibration should not be assumed and requires confirmation in a larger cohort.

**Conclusions:** BlazePose shows promising concurrent validity for dynamic flexion-focused tasks but exhibits clearly characterized failure modes—angle-dependent static bias, SLS instability, and session-to-session bias drift—that define the boundaries of its applicability in telerehabilitation. These preliminary boundaries may inform future protocol design and screening-level telerehabilitation use, pending larger-sample and clinical-population validation.

## Keywords

markerless motion capture; pose estimation; MediaPipe BlazePose; knee kinematics; concurrent

validity; inter-rater reliability; test–retest reliability; telerehabilitation; Kinovea; goniometry; digital health

## 1 Introduction

Accurate measurement of knee joint kinematics is fundamental to the assessment and management of musculoskeletal conditions, including anterior cruciate ligament (ACL) reconstruction recovery and knee osteoarthritis (OA) rehabilitation. Clinicians routinely monitor knee flexion and extension range of motion (ROM) to guide treatment progression, set discharge criteria, and evaluate surgical outcomes. In conventional clinical practice, the universal goniometer remains the most widely used instrument for static ROM measurement, offering acceptable intra-rater reliability when performed by trained clinicians (Brosseau et al [1]). However, the goniometer requires in-person contact, is limited to static or quasi-static postures, and cannot capture the dynamic joint behavior that characterizes functional tasks such as squatting, sit-to-stand transfers, and single-leg balance—tasks that more closely reflect the demands of daily living and return-to-activity benchmarks.

Laboratory-based marker-based motion capture systems (e.g., Vicon, Qualisys) are considered the gold standard for three-dimensional joint kinematic analysis and can quantify dynamic knee angles with sub-degree precision [11]. Yet these systems require dedicated laboratory space, specialized hardware, reflective marker placement by trained personnel, and post-processing expertise, resulting in per-session costs that limit their use to research settings and specialized clinical facilities [11,12]. This cost-access barrier is particularly problematic for telerehabilitation, where the clinical goal is to monitor patient progress remotely between in-person visits—a context in which neither the goniometer (requires co-location) nor marker-based motion capture (requires laboratory infrastructure) is feasible.

The rapid development of markerless pose estimation technologies has created new possibilities for accessible, low-cost movement analysis. Deep learning-based frameworks can estimate joint positions from standard video without physical markers. Stenum et al [13] demonstrated the feasibility of two-dimensional pose estimation for automated gait analysis using a similar pipeline. Among single-camera two-dimensional approaches, Gupta et al. [2] compared MediaPipe Pose with Kinovea 2D video annotation for knee flexion during gait, reporting a Pearson correlation of 0.941 and a mean absolute error of 5.88°. Langley et al [3] validated the smartphone-based Deep Vision system against both Kinovea and 3D motion capture in patients with knee OA, finding limits of agreement within clinical thresholds for peak flexion and ROM. Koul et al [4] demonstrated that the DETROM AI video goniometer provides reliable static knee ROM measurement compared with traditional goniometry in healthy adults (ICC = 0.98 for flexion). A recent study applied MediaPipe to joint angle and moment estimation during sit-to-stand transfers, extending the scope of single-camera pose estimation to functional tasks (Multimedia Tools and Applications [5]).

Multi-camera and three-dimensional markerless systems have generally achieved closer agreement with marker-based references. Lima et al [6] reported knee flexion root mean square error (RMSE) of 2.9–13.6° for OpenCap (a dual-smartphone system with OpenSim reconstruction) during squatting, jumping, and side-stepping tasks, with moderate to good inter-session reliability (ICC = 0.60–0.93). Yoma et al [7] found between-day RMSD < 5° and mean ICC of 0.77–0.83 for Theia3D during single-leg squat and single-leg landing tasks using a multi-camera setup. These multi-camera results establish an accuracy ceiling against which single-camera approaches can be benchmarked.

Despite this growing body of evidence, two critical gaps limit the clinical translation [12] of markerless pose estimation for telerehabilitation. First, existing validation studies have typically evaluated a single functional task (e.g., gait only, squat only, or sit-to-stand only) rather than the range of tasks used in a typical clinical knee assessment. Because pose estimation accuracy is known

to vary with joint configuration, movement speed, and the degree of limb occlusion, single-task findings do not generalize reliably to other tasks. Second, most studies have examined only one dimension of measurement quality—typically concurrent validity—without simultaneously assessing inter-rater reliability of the reference standard, or test–retest reliability of the automated system across sessions. For a telerehabilitation tool intended for longitudinal patient monitoring, session-to-session reproducibility and the stability of the measurement bias are at least as important as single-session accuracy, yet these properties remain largely uncharacterized for open-source pose estimation pipelines.

To our knowledge, few studies have evaluated an open-source markerless pose estimation system across multiple clinically relevant functional tasks within a single participant cohort using a multidimensional validation framework aligned with COSMIN measurement quality standards [15] that includes concurrent validity against both a clinical instrument and an expert video-analysis reference, inter-rater reliability of the reference standard, and test–retest reliability of both the automated and reference systems.

The purpose of this study was therefore to provide a multidimensional validation of the MediaPipe BlazePose Heavy model for knee joint angle measurement in a cohort of healthy young adults. Specifically, we evaluated: (1) inter-rater reliability of Kinovea 2D video annotation as the dynamic reference standard; (2) static concurrent validity of BlazePose against the clinical goniometer across four target angles; (3) dynamic concurrent validity of BlazePose against Kinovea during Squat, Sit-to-Stand, and bilateral Single-Leg Stance tasks; and (4) test–retest reliability of both BlazePose and Kinovea across two measurement sessions. We hypothesized that BlazePose would demonstrate acceptable concurrent validity for dynamic flexion-focused tasks but that accuracy and reliability would vary across tasks and across the range of motion, reflecting task-specific and geometry-dependent sources of error. This work represents the first stage (Study 1) of a planned three-study validation pipeline, to be followed by comparison against marker-based 3D motion capture in a laboratory setting (Study 2) and validation in clinical populations including post-anterior cruciate ligament reconstruction, post-total knee arthroplasty, and knee osteoarthritis (Study 3).

## 2 Methods

### Study Design

This cross-sectional validation study evaluated the concurrent validity of smartphone-based MediaPipe BlazePose for measuring knee joint angles against two reference standards: a clinical goniometer for static positions and expert manual video annotation via Kinovea for dynamic functional tasks. Secondary aims included assessment of inter-rater reliability of the Kinovea reference method and test–retest reliability of BlazePose across sessions. The study employed a three-layer comparison framework: clinical goniometer (clinical standard) → Kinovea 2D video analysis (human expert annotation) → BlazePose (automated pose estimation), enabling characterization of measurement agreement at each level of automation.

### Participants

Fifteen healthy adults (8 male, 7 female; mean age  $20.8 \pm 0.6$  years, range 20–22; height  $171.1 \pm 9.4$  cm; weight  $70.6 \pm 15.8$  kg; BMI  $23.9 \pm 3.5$  kg/m<sup>2</sup>, range 18.7–31.3) were recruited via convenience sampling from a university community. Inclusion criteria were: age 18–45 years, no current lower-extremity pain, and ability to perform a full-depth squat independently. Exclusion criteria included: history of lower-extremity surgery, current lower-extremity injury, or any musculoskeletal condition limiting knee range of motion. The study was approved by the Academic Ethics and Integrity Review Committee of Shanghai Sanda University (approval no. 2026006). All participants provided



informed consent prior to data collection. The sample size of 15 was determined by feasibility considerations (participant availability and resource constraints within a single-semester data collection window) rather than an a priori power calculation; this is acknowledged as a limitation, and the achieved precision of ICC estimates is reported via 95% confidence intervals throughout.

A subset of four participants (S01–S04) returned for a second session  $7 \pm 2$  days after their initial session to enable test–retest reliability analysis. The retest session followed an identical protocol (static and dynamic tasks). In total, the dataset comprised 15 initial datasets plus 4 retest datasets, yielding 342 trials processed through the BlazePose pipeline (228 static + 114 dynamic). The primary cross-sectional static validation used the 180 trials from S01–S15 (Session 1 only); the 48 retest static trials were reserved exclusively for the test–retest analysis.

## Equipment and Setup

All videos were recorded using a smartphone (iPhone 17 Pro, single rear camera) mounted on a tripod, positioned in the sagittal plane (lateral view) at knee height, 2.0 m from the participant. Recording parameters were standardized at  $1920 \times 1080$  resolution using H.264 encoding; participants S03–S15 and all retest sessions were recorded at 60 fps (~23 Mbps), while S01–S02 were recorded at 120 fps (~43 Mbps) during the pilot phase. Codec profile (High), pixel format (yuv420p), and color space (BT.709) were verified to be identical across all recordings via FFprobe. HDR was disabled and auto-exposure/auto-focus were locked prior to each recording. Floor tape marked both the participant standing position and the camera position to maintain consistent geometry across sessions.

Three small reflective markers were placed on each participant's test limb to support manual Kinovea annotation: the greater trochanter (hip), the lateral femoral epicondyle (knee), and the lateral malleolus (ankle). The BlazePose pipeline used MediaPipe's native keypoint detection and did not rely on these external markers. An ASIS marker was placed during some pilot trials but was not used in the final analysis pipeline. Participants wore dark-colored tight-fitting trousers or compression pants, a short-cropped top, and were barefoot. The background was a plain pale wall with even lighting and no harsh shadows.

## Data Collection Protocol

### Static Validation

For static validation, participants were seated with the knee positioned at four target flexion angles: 30°, 60°, 90°, and 120°, as set and measured by a standard clinical goniometer (fulcrum at the lateral femoral epicondyle, stationary arm aligned with the greater trochanter, moving arm aligned with the lateral malleolus). Each angle was held for approximately 5 seconds while video was recorded, and the procedure was repeated three times per angle, yielding 12 static trials per participant. The goniometer reading was recorded by the experimenter at the time of positioning and was removed before video capture began (i.e., the goniometer was not visible in the recorded video). This produced 180 static trials across the 15 cross-sectional participants ( $15 \times 4$  angles  $\times 3$  repetitions). An additional 48 static trials were collected from the four retest participants (S01–S04) using an identical protocol.

### Dynamic Validation

Three functional tasks were selected to represent a range of clinically relevant knee movements and to align with prior markerless motion capture validation literature [10,13]:

**Squat:** Five consecutive repetitions at a self-selected pace, with feet shoulder-width apart and arms extended forward for balance. The squat task provided the deepest knee flexion angles (~50–70°) and a clearly identifiable peak flexion frame.

**Sit-to-Stand (STS):** Five consecutive sit-to-stand transitions from a standard-height chair (approximately 45 cm seat height) at a self-selected pace, with arms extended forward for balance.

**Single-Leg Stance (SLS):** Standing on one leg for 15 seconds per trial. Two trials were performed per leg (left and right), yielding four SLS trials per participant. The supporting (standing) leg was the annotated limb for both BlazePose and Kinovea measurements. Beginning with S03, the protocol required that the non-stance leg remain out of the camera's field of view to prevent BlazePose tracking confusion between limbs.

Each participant completed six dynamic videos (1 Squat video containing 5 repetitions + 1 STS video containing 5 repetitions + 4 SLS videos). For each dynamic video, Kinovea raters identified the task-specific target frame — peak knee flexion (frame of minimum hip–knee–ankle angle) for Squat and Sit-to-Stand, and a canonical mid-stance frame within the steady-state single-leg stance for SLS — producing 14 Kinovea-annotated target-frame angles per participant (5 Squat + 5 STS + 2 SLS\_L + 2 SLS\_R). For Squat and STS the outcome is therefore the angle at peak knee flexion; for SLS the outcome is the steady-state supporting-leg knee angle, which in healthy participants lies near full extension (160–180°). Across 15 participants, this yielded 210 paired Kinovea annotations. The right knee was the default measurement side for all bilateral tasks (Squat, STS, SLS\_R); SLS\_L trials measured the left knee.

## Automated Pose Estimation: MediaPipe BlazePose

All videos were processed using MediaPipe BlazePose via the PoseLandmarker Task API (mediapipe 0.10.33, Python 3.9.6) with the Heavy model (pose\_landmarker\_heavy.task) [14], which provides the highest accuracy among the three available model tiers (Lite, Full, Heavy) and corresponds to model\_complexity = 2 in the legacy API. The model outputs 33 body landmarks per frame. Processing was configured in VIDEO running mode (frame-by-frame with temporal tracking) with a single-pose constraint (num\_poses = 1) and a default minimum pose-detection confidence of 0.5. All processing was performed on a local workstation (Apple M2 Max).

For each detected frame, the knee joint angle was computed as the included angle at the knee landmark formed by the hip–knee–ankle segment, using a standard two-dimensional vector angle calculation in pixel coordinates. For the right knee, landmarks 24 (RIGHT\_HIP), 26 (RIGHT\_KNEE), and 28 (RIGHT\_ANKLE) were used; for the left knee, landmarks 23, 25, and 27 respectively. The angle convention was set such that full knee extension corresponds to approximately 180° and deep flexion to approximately 50–70°. Per-frame confidence was defined as the minimum visibility score across the three landmarks (hip, knee, ankle), providing a conservative estimate of detection quality.

## Pose Estimation and Frame Retention

Pose landmarks were extracted using MediaPipe BlazePose (Heavy model) with a default minimum pose-detection confidence of 0.5. For five trials where the default threshold yielded no or near-zero valid detections (S09\_static\_120deg\_3, S10\_static\_90deg\_2, S10\_static\_120deg\_3, S04\_retest\_static\_120deg\_2, S04\_retest\_static\_120deg\_3), the threshold was lowered to 0.3 and the videos were reprocessed; detection succeeded in all five cases, with reprocessed trial-mean angles agreeing with adjacent same-condition trials to within 1°. All 180 static trials in the primary cross-sectional analysis (15 subjects × 12 conditions) were retained; the 48 Session-2 static trials of the test–retest sub-cohort were processed identically and used for the test–retest analysis.

Four trials from S09 and S10 yielded substantially fewer valid frames than the remaining dataset: S09\_static\_120deg\_3 (16/626 frames, 2.6%), S10\_static\_90deg\_3 (60/499, 12.0%), S10\_static\_90deg\_2 (123/569, 21.6%), and S10\_static\_120deg\_3 (129/604, 21.4%). The frames retained in these trials exhibited per-frame detection confidence values consistent with the broader

dataset (0.45–0.89), and their trial-mean knee-angle estimates agreed with the two adjacent repetitions from the same subject–angle cell to within  $1^\circ$  (maximum  $|\Delta| = 0.92^\circ$ ). These trials were therefore included in all statistical analyses without weighting or exclusion. Their reduced frame coverage is part of a between-batch detection-confidence shift observed in S07–S10.

Recording parameters were verified to be consistent across the dataset: all videos were captured at  $1920 \times 1080$  resolution using H.264 encoding, with S03–S15 recorded at 60 fps ( $\approx 23$  Mbps) and S01–S02 at 120 fps ( $\approx 43$  Mbps); codec profile, pixel format (yuv420p), and color space (BT.709) were identical throughout. The confidence shift observed in S07–S10 therefore cannot be attributed to recording-parameter variation, and its specific cause could not be identified in the present study (see Discussion).

## Peak Angle Extraction

For dynamic trials, peak knee flexion was defined as the minimum knee angle across all frames satisfying an anatomical plausibility constraint of  $\geq 10^\circ$ . Frames yielding knee angles below  $10^\circ$  were excluded from peak identification, as such values are anatomically implausible for knee flexion and indicative of transient pose-estimation failure (landmark degeneracy) rather than true joint motion. This constraint was applied uniformly to all dynamic and static trials; no trial was affected at the filtered-mean level (which additionally requires per-frame confidence  $\geq 0.8$ ).

## Confidence-Based Frame Filtering

For static trials, only frames with per-frame confidence  $\geq 0.8$  contributed to the filtered trial-mean angle. For the dynamic BlazePose-versus-Kinovea concurrent validity analysis, the BlazePose angle at the Kinovea-identified time stamp was treated as missing if the per-frame confidence at that frame fell below 0.3, interpreted as transient pose-estimation failure rather than a reliable estimate. This confidence filter excluded 33 of 210 Kinovea-paired observations (15.7%); the remaining 177 observations were used for concurrent-validity statistics.

## Reference Standard: Kinovea Manual Annotation

For dynamic trials, task-specific target-frame knee angles were independently annotated by two trained raters (Rater A and Rater B) using Kinovea 2025.2, an open-source 2D video analysis tool widely used in biomechanics and rehabilitation research. For each dynamic trial, the rater scrubbed through the video to identify the frame corresponding to peak knee flexion (the moment of minimum hip–knee–ankle angle for Squat and STS; for SLS, a canonical mid-stance frame within the steady-state single-leg stance). The rater then digitized three anatomical landmarks in the sagittal plane using the reflective markers as guides: the greater trochanter (hip), the lateral femoral epicondyle (knee), and the lateral malleolus (ankle). The resulting hip–knee–ankle angle was recorded along with the corresponding video timestamp.

Raters were blinded to each other's annotations and to the BlazePose output. The mean of the two raters' peak angle measurements served as the reference value for concurrent validity analysis. The mean of the two raters' peak-frame timestamps was used to determine the time-matched BlazePose angle for each observation (see below).

## BlazePose–Kinovea Time Matching

For the dynamic concurrent validity analysis, the BlazePose angle used for comparison was the time-matched value rather than BlazePose's own global peak. For each of the 210 Kinovea-annotated observations, the mean of the two raters' peak-frame timestamps was computed, and the BlazePose per-frame CSV was searched for the frame whose timestamp was closest to that mean. The BlazePose knee angle and confidence at that frame were recorded as the paired observation. This approach isolates pose-estimation accuracy from peak-detection differences between methods.

## Data Quality Grading

Each trial was assigned a quality grade (A–D) based on the proportion of frames exceeding a confidence threshold of 0.8: Grade A ( $\geq 80\%$  of frames), Grade B (50–79%), Grade C (20–49%), and Grade D ( $< 20\%$ ). This grading served as an internal quality-assurance metric and did not influence inclusion or exclusion decisions. Per-subject and per-task confidence distributions are reported in Supplementary Table S1 to enable readers to assess the between-session detection-confidence variability observed in the dataset (S01–S06: median trial confidence 0.94; S07–S10: median 0.48; S11–S14: median 0.91).

## Statistical Analysis

All statistical analyses were performed using Python 3.9.6 with the pingouin package for intraclass correlation coefficients and scipy.stats for inferential statistics. Figures were generated using matplotlib.

## Concurrent Validity

**Static (BlazePose vs. goniometer) and dynamic (BlazePose vs. Kinovea):** Agreement between BlazePose and the reference standard was assessed using a two-way random-effects, single-measures, absolute-agreement intraclass correlation coefficient, ICC(2,1), with 95% confidence intervals. Root mean square error (RMSE) and mean absolute error (MAE) were computed to quantify the magnitude of disagreement. Bland–Altman analysis [9] was performed to characterize systematic bias (mean difference between BlazePose and reference) and 95% limits of agreement ( $\text{LoA} = \text{bias} \pm 1.96 \times \text{SD}$ ). Pearson correlation coefficient ( $r$ ) was also computed; overall values are reported in the narrative text, with per-task RMSE and Pearson  $r$  available from the corresponding author upon request.

To address potential pseudoreplication arising from multiple trials per participant, we additionally computed all ICC analyses on subject-level (and where applicable subject-by-task-level) means as a sensitivity analysis; these subject-level estimates are reported in Supplementary Table S3.

For the static analysis, BlazePose filtered-mean knee angle (frames with confidence  $\geq 0.8$ ) was compared against the goniometer reading. Results were stratified by target angle ( $30^\circ$ ,  $60^\circ$ ,  $90^\circ$ ,  $120^\circ$ ) to assess angle-dependent accuracy patterns. For the dynamic analysis, the time-matched BlazePose angle was compared against the mean Kinovea angle from both raters. Results were reported both overall and stratified by task (Squat, STS, SLS).

## Inter-Rater Reliability

Inter-rater reliability of the Kinovea reference method was assessed by comparing peak knee flexion angles measured independently by Rater A and Rater B across all 210 dynamic trial annotations. ICC(2,1) with 95% CI, Bland–Altman bias and LoA, MAE, RMSE, and Pearson  $r$  were computed both overall and stratified by task. For the task-stratified analysis, differences in absolute inter-rater disagreement were tested using a Kruskal–Wallis omnibus test, followed by Mann–Whitney U pairwise comparisons. A minimum ICC of 0.90 was set a priori as the threshold for acceptable inter-rater agreement.

## Test–Retest Reliability

Test–retest reliability was assessed in four participants (S01–S04) who completed a second identical data collection session  $7 \pm 2$  days after their first. The retest analysis comprised four pre-specified components, each addressing a distinct aspect of session-to-session reproducibility.

First, BlazePose self-consistency was evaluated by paired comparison of the BlazePose filtered-mean angle (static) and time-matched peak flexion angle (dynamic) across Session 1 versus Session 2 ( $n =$

48 paired static observations and  $n = 24$  paired dynamic observations).

Second, Kinovea reference stability was evaluated by paired comparison of the Kinovea two-rater mean angles across Session 1 versus Session 2 in the same four participants ( $n = 56$  paired observations: 4 subjects  $\times$  14 trials), to verify that the reference standard itself was reproducible across sessions.

Third, concurrent validity replication in Session 2 was evaluated by recomputing the BlazePose-vs-Kinovea concurrent validity statistics on the Session-2 data alone ( $n = 35$  paired observations after applying the same  $\text{bp\_confidence} \geq 0.3$  filter used in the primary analysis), to test whether the Session-1 concurrent validity finding generalized to a follow-up session.

Fourth, bias stability across sessions was evaluated by computing the per-trial BlazePose-minus-Kinovea difference in each session and treating these differences as paired observations across sessions ( $n = 34$  trials with valid measurements in both sessions). The ICC of these between-method differences across sessions tests whether a fixed offset correction would generalize across sessions for the same participant — a clinically relevant question for longitudinal telerehabilitation monitoring.

All four components used ICC(2,1) (two-way random-effects, single-rater, absolute agreement), Bland–Altman bias and 95% LoA, MAE, and RMSE. Results are reported overall, per-subject, and stratified by task (Squat, Sit-to-Stand, Single-Leg Stance) where sample sizes permitted.

## Interpretation Thresholds

ICC values were interpreted according to Koo and Li [8]:  $< 0.50$  = poor,  $0.50$ – $0.75$  = moderate,  $0.75$ – $0.90$  = good, and  $> 0.90$  = excellent reliability. An RMSE below  $5^\circ$  was considered clinically acceptable based on prior markerless motion capture validation literature. Statistical significance was set at  $\alpha = 0.05$  for all inferential tests.

**Table 1.** Summary of validation analyses, sample sizes, and outcome measures.

| Analysis                    | Comparison                               | n       | Subjects | Primary Metrics                        |
|-----------------------------|------------------------------------------|---------|----------|----------------------------------------|
| Static validity             | BlazePose vs. Goniometer                 | 180     | S01–S15  | ICC(2,1), RMSE, Bland–Altman           |
| Dynamic validity            | BlazePose vs. Kinovea (mean of 2 raters) | 177*    | S01–S15  | ICC(2,1), RMSE, Bland–Altman           |
| Inter-rater reliability     | Rater A vs. Rater B (Kinovea)            | 210     | S01–S15  | ICC(2,1), Bland–Altman, Kruskal–Wallis |
| Test–retest 4A: BP self     | Session 1 vs. Session 2 (BlazePose)      | 48 / 24 | S01–S04  | ICC(2,1), Bland–Altman, MAE            |
| Test–retest 4B: Kin self    | Session 1 vs. Session 2 (Kinovea mean)   | 56      | S01–S04  | ICC(2,1), Bland–Altman, MAE            |
| Test–retest 4C: Validity S2 | BlazePose vs. Kinovea (Session 2)        | 35**    | S01–S04  | ICC(2,1), RMSE, Bland–Altman           |
| Test–retest 4D: Bias stab.  | (BP – Kin)_S1 vs (BP – Kin)_S2           | 34      | S01–S04  | ICC(2,1)                               |

\*33 of 210 trials excluded due to BlazePose confidence  $< 0.3$  at the Kinovea-identified peak frame. \*\*21 of 56 trials excluded due to BlazePose

confidence < 0.3 (concentrated in S01 retest cohort). BP = BlazePose; Kin = Kinovea.

### 3 Results

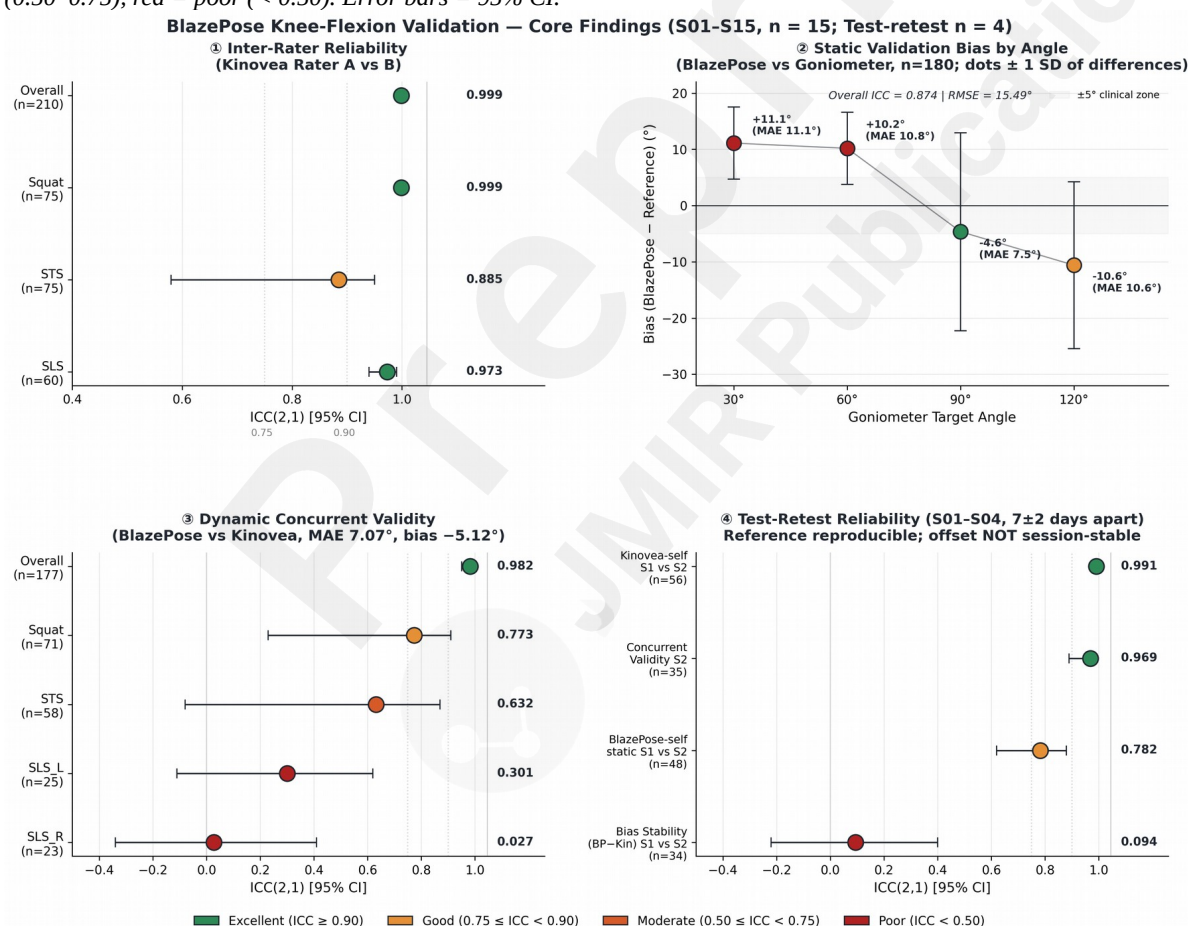
#### Participants

Fifteen healthy adults (8 male, 7 female) participated in the cross-sectional analysis (mean age  $20.8 \pm 0.6$  years, range 20–22; height  $171.1 \pm 9.4$  cm; weight  $70.6 \pm 15.8$  kg; BMI  $23.9 \pm 3.5$  kg/m<sup>2</sup>, range 18.7–31.3). All 15 participants completed the full data collection protocol without adverse events. Four participants (S01–S04; 2 male, 2 female) returned for the retest session within the planned  $7 \pm 2$ -day window.

The dataset comprised 342 trials processed through the BlazePose pipeline (228 static + 114 dynamic). For the primary cross-sectional analyses, 180 static trials (15 subjects  $\times$  12 conditions) and 210 dynamic Kinovea-annotated observations were available.

An overview of the four primary analyses is provided in Figure 1; detailed numerical results are reported in Tables 1–5.

**Figure 1.** Overview of the four core findings. Panel ① inter-rater reliability of Kinovea (Rater A vs. B) by task. Panel ② angle-dependent bias of BlazePose static measurements. Panel ③ BlazePose-vs-Kinovea concurrent validity by task. Panel ④ test–retest reliability across four sub-analyses. Color coding: green = excellent ( $ICC \geq 0.90$ ), amber = good ( $0.75–0.90$ ), orange = moderate ( $0.50–0.75$ ), red = poor ( $< 0.50$ ). Error bars = 95% CI.



#### Inter-Rater Reliability of Kinovea Annotations

Inter-rater reliability of the Kinovea reference standard was excellent overall ( $ICC(2,1) = 0.999$  [95% CI: 0.998, 0.999], MAE =  $1.56^\circ$ , n = 210), supporting its use as the reference for the dynamic concurrent validity analysis (Table 2).

When stratified by task, inter-rater agreement was highest for Squat (ICC = 0.999, MAE = 0.39°), followed by SLS (ICC = 0.973, MAE = 1.05°). Sit-to-Stand exhibited the lowest inter-rater reliability (ICC = 0.885 [0.58, 0.95], MAE = 3.13°), with a systematic bias of -2.65° (Rater A consistently lower than Rater B). The between-task difference in absolute inter-rater disagreement was significant (Kruskal–Wallis  $H = 71.5$ ,  $p < 0.001$ ), with all pairwise comparisons reaching significance (Mann–Whitney  $U$ ,  $p < 0.001$ ). This task-dependent pattern likely reflects the difficulty of identifying the precise frame of peak flexion in STS, where the transition from seated to standing involves a gradual angular change, compared with the clearly defined lowest point in Squat (Figure 1, panel ①; Figure 2C).

**Table 2. Inter-rater reliability of Kinovea peak knee-flexion annotations ( $N = 210$ ).**

| Group   | n   | Bias   | 95%<br>LoA        | MAE   | RMSE  | r     | ICC<br>[95%<br>CI]         |
|---------|-----|--------|-------------------|-------|-------|-------|----------------------------|
| Overall | 210 | -1.15° | [-5.79,<br>+3.49] | 1.56° | 2.63° | 0.999 | 0.999<br>[0.998,<br>0.999] |
| Squat   | 75  | -0.09° | [-1.43,<br>+1.24] | 0.39° | 0.68° | 0.999 | 0.999<br>[0.999,<br>0.999] |
| STS     | 75  | -2.65° | [-8.98,<br>+3.67] | 3.13° | 4.16° | 0.936 | 0.885<br>[0.58,<br>0.95]   |
| SLS     | 60  | -0.58° | [-3.04,<br>+1.87] | 1.05° | 1.37° | 0.978 | 0.973<br>[0.94,<br>0.99]   |

Bias = mean(Rater A - Rater B). ICC = ICC(2,1), two-way random-effects, single-rater, absolute agreement. Kruskal–Wallis between-task  $|A - B|$ :  $H = 71.5$ ,  $p < 0.001$ .

## Static Concurrent Validity: BlazePose vs. Goniometer

Overall agreement between BlazePose and the clinical goniometer across 180 static trials was good (ICC(2,1) = 0.874 [0.84, 0.91]), though RMSE (15.49°) substantially exceeded the 5° clinical equivalence threshold (Table 3).

Angle-stratified analysis revealed a systematic, direction-reversing bias pattern. At shallow goniometer-set flexion (30° and 60°), BlazePose overestimated the corresponding extension angle by approximately +10–11° (30°: bias = +11.11°, MAE = 11.11°; 60°: bias = +10.18°, MAE = 10.82°). At deep flexion (120°), BlazePose underestimated by a similar magnitude (bias = -10.60°, MAE = 10.60°). The 90° condition served as the crossover point, with near-zero mean bias (-4.64°) but substantially wider limits of agreement (LoA: [-39.15°, +29.86°]). Across the four target conditions, BlazePose extension estimates regressed toward approximately 105° (corresponding to ~75° flexion — the midpoint of the tested range). This regression-toward-the-mean pattern is geometrically consistent with planar landmark localization at near-collinear hip–knee–ankle configurations: small absolute pixel errors in landmark placement translate into proportionally larger angular errors when the segments are close to alignment, producing the observed compression of the angle range (further discussed in Discussion) (Figure 1, panel ②; Figure 2A).

**Table 3. Static concurrent validity: BlazePose vs. goniometer ( $N = 180$ , S01–S15).**

| Group          | n   | Bias    | 95%<br>LoA         | MAE    | RMSE   | SD(diff) | SEM   |
|----------------|-----|---------|--------------------|--------|--------|----------|-------|
| Overall        | 180 | —       | —                  | 9.99°  | 15.49° | —        | —     |
| 30° (ext 150°) | 45  | +11.11° | [-1.45,<br>+23.67] | 11.11° | 12.67° | 6.41°    | 4.53° |
| 60° (ext 120°) | 45  | +10.18° | [-2.37,<br>+12.73] | 10.82° | 11.99° | 6.40°    | 4.53° |

|             |  |         |          |        |        |        |        |
|-------------|--|---------|----------|--------|--------|--------|--------|
| 120°)       |  |         | +22.73]  |        |        |        |        |
| 90° (ext 45 |  | -4.64°  | [-39.15, | 7.45°  | 18.02° | 17.60° | 12.45° |
| 90°)        |  |         | +29.86]  |        |        |        |        |
| 120° 45     |  | -10.60° | [-39.69, | 10.60° | 18.10° | 14.84° | 10.49° |
| (ext 60°)   |  |         | +18.48]  |        |        |        |        |

Bias = BlazePose extension angle – expected extension angle (180° – goniometer flexion). Overall ICC(2,1) [95% CI] = 0.874 [0.84, 0.91]. SEM = SD(diff) /  $\sqrt{2}$ .

## Dynamic Concurrent Validity: BlazePose vs. Kinovea

Of 210 dynamic Kinovea-annotated observations, 33 (15.7%) were excluded due to BlazePose per-frame confidence below 0.3 at the Kinovea-identified peak frame, leaving 177 paired observations for the concurrent validity analysis. Excluded trials were concentrated in participants S07–S10, consistent with the between-batch confidence shift identified during data quality assessment.

The Session-2 exclusion rate (21 of 56 trials, 37.5%) was substantially higher than the Session-1 rate (33 of 210 trials, 15.7%), with the increase concentrated in S01\_retest (12 of 14 trials excluded). This differential exclusion pattern is itself a finding: BlazePose's reliability for an individual participant is not guaranteed to be replicated in a follow-up session, which has direct implications for longitudinal monitoring (see Discussion).

Overall concurrent validity was excellent (ICC(2,1) = 0.982 [0.95, 0.99], Pearson  $r$  = 0.987), with a systematic negative bias of  $-5.12^\circ$  indicating that BlazePose underestimated the task-specific target angle (peak flexion for Squat and STS; steady-state supporting-leg knee angle for SLS) relative to Kinovea (Table 4). The 95% limits of agreement were  $[-21.39^\circ, +11.15^\circ]$ , and MAE was  $7.07^\circ$ .

Task-stratified results revealed a clear hierarchy. Squat showed strong relative agreement (Pearson  $r$  = 0.885) but moderate absolute agreement (ICC(2,1) = 0.773). The gap between these two indices reflects the systematic  $\sim 6^\circ$  underestimation of peak flexion: BlazePose tracks the relative ranking of subjects' peak flexion well (Pearson  $r$ ), but the fixed downward offset reduces absolute agreement (ICC(2,1)). The same pattern, with a similar offset, was observed for STS (ICC = 0.632, MAE =  $6.68^\circ$ , bias =  $-6.43^\circ$ ), with tighter limits of agreement than Squat ( $[-14.42^\circ, +1.56^\circ]$ ). SLS showed poor agreement (combined ICC = 0.136, MAE =  $7.51^\circ$ ), with the per-side results similarly poor (SLS\_R ICC = 0.027, SLS\_L ICC = 0.301). The poor SLS performance reflects a near-extension floor effect — the supporting knee remains close to full extension ( $160$ – $180^\circ$ ) throughout the stance, where small absolute landmark displacements produce proportionally large angular errors (Figure 1, panel ③; Figure 2B).

Subject-level sensitivity analysis (averaging trial-level observations within each participant prior to ICC computation) yielded ICC estimates within 0.05 of the trial-level values across all overall and per-task comparisons (Supplementary Table S3), indicating that within-subject trial repetition did not materially alter the point estimates or interpretation of the primary trial-level ICC findings reported above.

**Table 4. Dynamic concurrent validity: BlazePose vs. Kinovea (N = 177/210, S01–S15).**

| Group   | n   | Bias          | 95% LoA            | MAE          | ICC [95% CI]        |
|---------|-----|---------------|--------------------|--------------|---------------------|
| Overall | 177 | $-5.12^\circ$ | $[-21.39, +11.15]$ | $7.07^\circ$ | 0.982 [0.95, 0.99]  |
| Squat   | 71  | $-6.17^\circ$ | $[-18.50, +6.17]$  | $7.10^\circ$ | 0.773 [0.23, 0.91]  |
| STS     | 58  | $-6.43^\circ$ | $[-14.42, +1.56]$  | $6.68^\circ$ | 0.632 [-0.08, 0.87] |
| SLS_L   | 25  | $+0.90^\circ$ | $[-20.46, +22.27]$ | $6.75^\circ$ | 0.301 [-0.11, 0.62] |



|           |    |        |                     |       |                           |
|-----------|----|--------|---------------------|-------|---------------------------|
| SLS_R     | 23 | -5.18° | [-33.11,<br>+22.76] | 8.34° | 0.027<br>[-0.34,<br>0.41] |
| SLS comb. | 48 | -2.01° | [-27.20,<br>+23.18] | 7.51° | 0.136<br>[-0.15,<br>0.40] |

*Bias = BlazePose – Kinovea (mean of two raters). Negative bias = BlazePose underestimates peak flexion. 33/210 trials excluded for per-frame confidence < 0.3.*

## Test–Retest Reliability

### Kinovea Reference Stability Across Sessions

Kinovea-self test–retest reliability was excellent overall (ICC(2,1) = 0.991 [0.98, 0.99], MAE = 5.63°, n = 56; Table 5), confirming the temporal stability of the reference standard. Per-subject ICCs were uniformly high (range: 0.989–0.994), indicating that raters reproduced their own peak-frame identifications with high fidelity across sessions. Task-stratified results showed adequate reliability for Squat (ICC = 0.873) and moderate reliability for STS (ICC = 0.631), but negative ICC values for both SLS\_L (-0.713) and SLS\_R (-0.511). The negative SLS ICCs were observed even for the Kinovea raters themselves, providing strong evidence that SLS measurement instability is task-intrinsic rather than method-specific.

### BlazePose Self-Consistency Across Sessions

BlazePose static test–retest reliability was good overall (ICC(2,1) = 0.782 [0.62, 0.88], n = 48), with substantial between-subject variability: S02 (ICC = 0.980), S03 (ICC = 0.997), and S04 (ICC = 0.947) showed excellent within-subject reproducibility, while S01 (ICC = 0.472) showed poor reproducibility with a large positive bias (S2 – S1 = +24.14°). Dynamic test–retest reliability was moderate overall (ICC = 0.522, n = 24), again driven by S01; S03 achieved near-perfect reproducibility (ICC = 0.998, MAE = 3.16°).

### Replication of Concurrent Validity in Session 2

The BlazePose-vs-Kinovea concurrent validity observed in Session 1 was replicated in Session 2 (n = 35 after exclusions; ICC = 0.969 [0.89, 0.99], MAE = 8.95°, bias = -8.10°; Table 5). The systematic underestimation of peak flexion relative to Kinovea persisted (Session 1: -5.12°, Session 2: -8.10°), and the task-stratified pattern was consistent: Squat ICC = 0.818, STS ICC = 0.535, SLS combined ICC near zero.

### Bias Stability Across Sessions

The per-trial BlazePose-minus-Kinovea difference was computed for each session and compared across sessions in 34 matched observations. The between-session ICC of these differences was 0.094 [-0.22, 0.40] (Table 5), indicating that the BlazePose-vs-Kinovea bias was not stable across sessions. For Squat, the most favorable task, the bias ICC was only 0.401. This finding indicates that a fixed offset correction (e.g., adding a constant to BlazePose readings) would not generalize across measurement sessions, which constitutes a key limitation for longitudinal telerehabilitation monitoring (Figure 1, panel ④). Although the 95% confidence interval for this ICC ([-0.22, 0.40]) does not exclude moderate stability, the point estimate and the clinical consequence of assuming stability when it may not exist argue for a conservative interpretation; future studies with larger retest cohorts should re-examine this question, including sensitivity analyses excluding outlier participants.

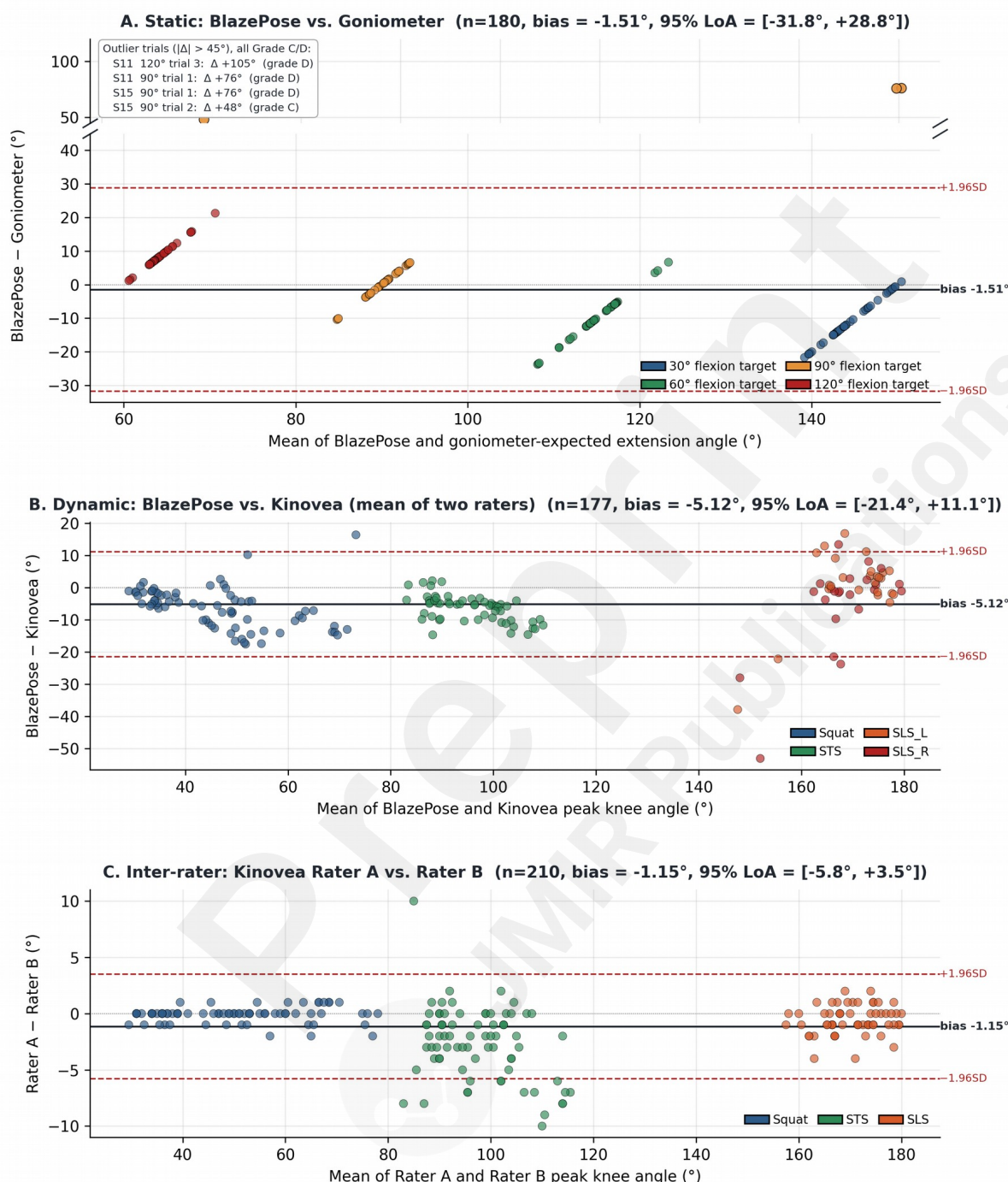
*Table 5. Test–retest reliability summary (S01–S04, Session 1 vs. Session 2, 7 ± 2 days apart).*

| Sub-analysis | n | Bias | 95% LoA | MAE | ICC(2,1)<br>[95% CI] |
|--------------|---|------|---------|-----|----------------------|
|--------------|---|------|---------|-----|----------------------|

|                                                     |    |         |                      |        |                           |
|-----------------------------------------------------|----|---------|----------------------|--------|---------------------------|
| 4A.<br>BlazePose-<br>self (static,<br>S1 vs S2)     | 48 | +7.31°  | [-28.04,<br>+42.65]  | 10.32° | 0.782 [0.62,<br>0.88]     |
| 4A.<br>BlazePose-<br>self<br>(dynamic,<br>S1 vs S2) | 24 | -18.92° | [-119.89,<br>+82.04] | 34.71° | 0.522 [0.17,<br>0.76]     |
| 4B.<br>Kinovea-<br>self (S1 vs<br>S2)               | 56 | -1.19°  | [-14.42,<br>+12.05]  | 5.63°  | 0.991 [0.98,<br>0.99]     |
| 4C.<br>BlazePose<br>vs Kinovea<br>(Session 2)       | 35 | -8.10°  | [-30.89,<br>+14.69]  | 8.95°  | 0.969 [0.89,<br>0.99]     |
| 4D. Bias<br>stability<br>(BP-Kin)<br>S1 vs S2       | 34 | -2.45°  | [-18.91,<br>+14.01]  | 5.19°  | 0.094<br>[-0.22,<br>0.40] |

*Bias direction: 4A and 4B use Session 2 – Session 1; 4C uses BlazePose – Kinovea (Session 2 only); 4D treats per-trial (BlazePose – Kinovea) differences as paired observations across sessions, so the ICC measures the reproducibility of the between-method difference. Per-subject and per-task breakdowns for all four sub-analyses are provided in Supplementary Table S2. The exploratory nature of the retest sub-sample (n = 4 participants) should be considered when interpreting these estimates.*

Figure 2. Bland–Altman plots for the three primary validation comparisons. (A) Static BlazePose vs. goniometer (n = 180; cross-sectional cohort), colored by target flexion angle. The vertical spread reflects the angle-dependent bias pattern (overestimation at 30°/60° flexion, underestimation at 120°). Panel A uses a broken y-axis: four trials (S11, S15; both Grade C/D) with differences exceeding 45° are shown in a separate zone above the axis break, while the main band of 176 trials is displayed at full resolution below. (B) Dynamic BlazePose vs. Kinovea (n = 177 after confidence ≥ 0.3 filter), colored by task; systematic ~5° underestimation. (C) Inter-rater Kinovea Rater A vs. Rater B (n = 210); tight agreement overall, with task-specific spread highest in STS. Solid line = bias; dashed lines = 95% limits of agreement.

**Figure 2. Bland-Altman plots for three primary validation comparisons**

## Data Quality Overview

Of 342 total trials, internal quality grading (A–D, based on percentage of frames with per-frame confidence  $\geq 0.8$ ) yielded 47.9% Grade A and 36.8% Grade D. The high Grade-D proportion was attributable to a between-batch detection-confidence shift: median trial confidence was 0.94 in S01–S06, 0.48 in S07–S10, and 0.91 in S11–S15. This shift could not be explained by recording-parameter variation (Methods §Equipment and Setup); its specific cause remains unidentified and is discussed in Discussion §Limitations.

## 4 Discussion

This study evaluated the concurrent validity, inter-rater reliability, and test–retest reliability of the open-source MediaPipe BlazePose pipeline for measuring knee joint angles across multiple functional tasks. Across 15 healthy adults (8 male, 7 female; age  $20.8 \pm 0.6$  years) and four retest participants, BlazePose achieved excellent overall concurrent validity against Kinovea expert annotations for task-specific dynamic knee-angle outcomes (peak knee flexion for Squat and Sit-to-Stand, and steady-state supporting-leg knee angle for SLS) (pooled ICC = 0.982 [0.95, 0.99], MAE =  $7.07^\circ$ ), although within-task ICCs were lower (Squat 0.773, STS 0.632, SLS 0.136) due to reduced between-subject variance within each task, with a systematic underestimation of approximately  $5\text{--}8^\circ$  relative to the reference. However, three findings circumscribe its clinical applicability for telerehabilitation: (1) angle-dependent bias in static measurement that reverses sign across the knee range of motion, (2) task-intrinsic instability of single-leg stance measurements observed for both BlazePose and the Kinovea reference itself, and (3) the absence of session-to-session bias stability (ICC = 0.094 [–0.22, 0.40]) that prevents reliable fixed-offset correction in longitudinal monitoring.

### 4.1 Reference Standard Reliability

The excellent inter-rater reliability of Kinovea peak-flexion annotation (overall ICC = 0.999 [0.998, 0.999]; per-task ICC range 0.885–0.999) and its high temporal stability across sessions (overall test–retest ICC = 0.991 [0.98, 0.99]) support its role as the reference standard for the dynamic concurrent validity analysis. Both findings are consistent with prior reports of Kinovea reliability in healthy populations and confirm that the systematic differences observed between BlazePose and Kinovea reflect properties of the BlazePose pipeline rather than rater inconsistency (Figure 1, panel ①).

### 4.2 Concurrent Validity in Context of Prior Literature

The overall dynamic concurrent validity of BlazePose against Kinovea peak knee flexion was excellent (ICC = 0.982 [0.95, 0.99], Pearson  $r = 0.987$ ,  $n = 177$ ), with a systematic underestimation of  $-5.12^\circ$ . Task-stratified analysis revealed strong agreement for Squat (Pearson  $r = 0.885$ ; ICC = 0.773 [0.23, 0.91]) and moderate agreement for Sit-to-Stand ( $r = 0.896$ ; ICC = 0.632 [–0.08, 0.87]), with the reduced ICC values relative to Pearson  $r$  reflecting the systematic  $\sim 6^\circ$  downward offset. The agreement we observed for Squat (MAE =  $7.10^\circ$ ) is comparable to Gupta et al.'s [2] MediaPipe-vs-Kinovea result for gait (MAE =  $5.88^\circ$ ), and to Lima et al.'s [6] OpenCap during squatting (knee RMSE =  $2.9\text{--}13.6^\circ$  vs. marker-based 3D motion capture). Bae et al [10] similarly reported excellent concurrent validity for a deep learning markerless system during overhead squats (ICC = 0.98), and our finding extends this prior single-task evidence by demonstrating that BlazePose's concurrent validity holds across multiple functional tasks within a single participant cohort (Figure 1, panel ③). The  $\sim 6^\circ$  systematic underestimation by BlazePose merits explicit attention because it persists across Squat and Sit-to-Stand, across Session 1 and Session 2, and across most participants. The gap between the overall ICC (0.982) and task-specific ICCs (Squat 0.773, STS 0.632) also deserves comment: the overall figure benefits from between-task variance in peak flexion magnitudes (Squat  $\sim 50^\circ$  flexion, STS  $\sim 95^\circ$ , SLS  $\sim 170^\circ$  extension), which inflates the ICC denominator. Within-task ICC is the more clinically meaningful metric because clinicians interpret measurements within task-specific reference ranges. The most plausible mechanism for the systematic offset, consistent with the static-bias analysis discussed below, is that BlazePose's hip and ankle landmarks systematically deviate from the rater-used anatomical reference points (greater trochanter, lateral malleolus). This consistent direction of offset suggests that, in principle, a calibration correction could be applied—a possibility we test in the bias-stability analysis and ultimately reject.

### 4.3 Static Angle-Dependent Bias

Static BlazePose measurements showed a systematic angle-dependent bias relative to the goniometer reference (overall ICC = 0.874 [0.84, 0.91], RMSE = 15.49°). At shallow goniometer-set flexion (30°, 60°), BlazePose overestimated the corresponding knee extension angle by approximately 10°; at deep flexion (120°), it underestimated by a similar magnitude, with a crossover near 90° target flexion. This direction-reversing pattern indicates that BlazePose static measurements are subject to geometrically mediated noise rather than a fixed scaling error. Koul et al [4] reported broadly feasible static knee ROM measurement using the DETROM AI video goniometer in healthy adults (ICC = 0.98), and Langley et al [3] found 95% limits of agreement exceeding  $\pm 10^\circ$  at peak flexion for Kinovea in knee-OA patients—a pattern consistent with our angle-dependent findings (Figure 1, panel ②).

The geometric origin of this pattern lies in the relationship between landmark localization error in pixel space and the resulting angular error at the knee joint. The MediaPipe BlazePose hip and ankle keypoints localize to anatomical positions that do not exactly match the clinical reference points used by the goniometer rater; inspection of pose-overlay videos showed BlazePose's hip keypoint located anterior to the greater trochanter (near the inguinal fold). Small displacements in pixel space at these landmarks produce small angular errors at deep flexion (where the hip–knee–ankle vectors form a sharp angle) but disproportionately large angular errors at near-extension (where the vectors approach collinearity). The result is a regression toward an intermediate value approximately equal to the midpoint of the tested range. Because this geometric amplification is participant-specific—different leg-segment proportions and body composition lead to different landmark-location offsets—the magnitude of the bias varies across individuals, as observed in the wide between-subject variance at the same target angle.

### 4.4 Single-Leg Stance: Task-Intrinsic Instability

BlazePose performed poorly for single-leg stance (combined ICC = 0.136 [−0.15, 0.40]; per-side ICCs near zero), but this finding requires careful interpretation. The Kinovea reference itself showed poor test–retest reproducibility for SLS (SLS\_L ICC = −0.713, SLS\_R = −0.511), despite per-subject Kinovea reliability  $\geq 0.989$ . The combination of poor concurrent validity and poor reference-standard test–retest reliability indicates that SLS measurement instability is task-intrinsic rather than method-specific. Two factors contribute: first, the supporting knee remains near full extension (160–180°) throughout the stance, where small landmark-localization errors are geometrically amplified into large angular errors regardless of method; second, the swinging leg can occlude the supporting leg's lower-extremity landmarks in single-camera sagittal footage—an issue absent in multi-camera setups such as Theia3D, which reports between-day RMSD  $< 5^\circ$  for similar tasks (Yoma et al [7]). We therefore recommend against using single-camera BlazePose for SLS knee-flexion measurement in telerehabilitation contexts; if SLS assessment is clinically required, a multi-view setup or a different outcome metric (e.g., center-of-mass sway) should be considered.

### 4.5 Bias Instability: Implications for Telerehabilitation

In an exploratory analysis with a small retest subsample ( $n = 4$  participants), the BlazePose-vs-Kinovea per-trial bias was not stable across sessions (ICC = 0.094 [−0.22, 0.40],  $n = 34$ ). For Squat—the task where BlazePose performed best—between-session bias-stability ICC was only 0.401 [−0.06, 0.72]. This finding has direct implications for telerehabilitation deployment. A common engineering response to systematic bias is to apply a fixed offset correction (e.g., adding  $+5^\circ$  to BlazePose readings to account for the observed underestimation). Our results indicate this strategy would not generalize across measurement sessions for the same patient: the per-trial bias in Session 1 does not predict the per-trial bias in Session 2. The mechanism is not fully elucidated; possibilities include subtle lighting or clothing changes, BlazePose's known sensitivity to small image-

distribution shifts, or session-specific variation in marker placement and participant posture. This means that any clinical interpretation of BlazePose readings in longitudinal monitoring should not assume a stable calibration; instead, each session must be treated as an independent measurement, ideally accompanied by an in-person reference measurement at clinically relevant intervals (Figure 1, panel ④).

Sensitivity analysis excluding S01\_retest (which contributed 5 of 34 paired differences and was the participant with the highest Session-2 confidence-filter exclusion rate) yielded an overall bias-stability ICC of 0.137 [−0.18, 0.45] ( $n = 29$ ), supporting the interpretation that the poor between-session reproducibility of the BlazePose-vs-Kinovea bias is not an artefact of the S01\_retest outlier and holds in the remaining three retest participants.

In our small retest cohort, one of four participants (S01) showed markedly degraded BlazePose performance in Session 2, with 12 of 14 trials excluded for low confidence and substantially worse test–retest reliability (static ICC = 0.472 [−0.06, 0.81], MAE = 27.14°) compared with S02–S04 (static ICC = 0.95–0.997). The mechanism could not be identified despite consistent recording parameters (verified by FFprobe) and identical protocols. Although  $n = 4$  is far too small to generalize, even a single such session-level breakdown in a clinical population would render the monitoring data unusable. This argues for quality-assurance gates in any telerehabilitation deployment—for example, a clinician spot-checking pose-overlay video before clinical interpretation.

## 4.6 Limitations

Several limitations should be considered when interpreting these results. First, the sample was small ( $n = 15$  cross-sectional,  $n = 4$  retest) and comprised healthy young adults (age 20–22 years) recruited at a single site with a single recording device. Statistical power for detecting subgroup differences was limited, and the 95% confidence intervals for some per-angle and per-task ICCs were wide. The findings may not generalize to clinical populations (e.g., post-ACL reconstruction, post-total knee arthroplasty, or knee osteoarthritis), who may exhibit different movement patterns and compensatory strategies that challenge pose estimation differently.

Second, we used the clinical goniometer (static) and Kinovea (dynamic) as reference standards rather than marker-based 3D motion capture, which is the absolute gold standard for joint kinematics. The systematic  $\sim 6^\circ$  underestimation we attribute to BlazePose could be partially confounded by Kinovea's own residual error relative to a 3D reference. Third, only the BlazePose Heavy model variant was tested; the Lite and Full models may exhibit different bias profiles. Fourth, reflective markers were placed on anatomical landmarks for Kinovea visibility; although we expect minimal effect given their small size ( $\sim 10$  mm diameter), a controlled marker-on/marker-off comparison was not performed. Fifth, five static trials required reprocessing at a lowered pose-detection confidence threshold (0.3 instead of the 0.5 default) due to initial detection failure (see Methods); reprocessed trial-mean angles agreed with adjacent same-condition trials to within  $1^\circ$ , supporting the validity of the rescued measurements, but their inclusion does represent a procedural deviation from the uniform pipeline applied to the remaining trials. Finally, our validation was specific to sagittal-plane peak knee flexion during four functional tasks; frontal-plane and transverse-plane angles, as well as temporal kinematic variables (e.g., angular velocity, time to peak flexion), were not assessed. Sixth, the confidence-based exclusion filter removed 33 of 210 dynamic observations (15.7%), with exclusions concentrated in participants S07–S10; because the excluded observations may represent systematically poorer BlazePose performance, the reported concurrent validity estimates should be interpreted as conditional on adequate pose-detection confidence rather than as unconditional accuracy. Seventh, over 60 ICC estimates were computed across task-stratified and angle-stratified analyses without formal correction for multiple comparisons; these stratified estimates should be considered exploratory, and the overall (unstratified) ICCs remain the primary outcomes.

## 4.7 Clinical Implications and Future Directions

Despite these limitations, the present findings provide preliminary evidence supporting a potential role for single-camera BlazePose in telerehabilitation: approximate screening-level monitoring of dynamic flexion-focused tasks (Squat, Sit-to-Stand) between in-person visits, with the understanding that measurements carry a systematic  $\sim 6^\circ$  underestimation and session-to-session variability that precludes precise longitudinal tracking. BlazePose is not a replacement for in-person goniometry; rather, its value lies in the cost-zero, open-source accessibility of the MediaPipe pipeline, which requires only a standard smartphone or webcam.

Several directions follow from this work. First, a larger healthy-adult cohort ( $n \geq 30$ ) and multi-site replication would narrow confidence intervals and test generalization across recording environments and devices. Second, validation against marker-based 3D motion capture (e.g., Qualisys, Vicon) would establish absolute accuracy benchmarks beyond the Kinovea-relative values reported here. Third, clinical population validation—particularly in patients with restricted range of motion or atypical gait patterns—is essential before telerehabilitation deployment. Fourth, the confidence-score paradox observed in our data (one participant showed low confidence but high accuracy) warrants dedicated investigation of BlazePose's per-frame confidence as a quality-assurance metric. Finally, setup standardization protocols for telerehabilitation (camera height, distance, background, attire) should be developed and tested to minimize the session-to-session variability documented here. These directions form a coherent validation pipeline (Study 2: 3D motion-capture benchmarking; Study 3: clinical populations) within which the present work establishes the open-source single-camera baseline (Study 1).

## 4.8 Conclusions

MediaPipe BlazePose demonstrates promising concurrent validity for research and screening purposes when measuring task-specific dynamic knee-angle outcomes, particularly peak knee flexion during Squat and Sit-to-Stand tasks, with a systematic  $\sim 6^\circ$  underestimation relative to Kinovea expert annotation. Static accuracy is angle-dependent and exceeds the  $5^\circ$  clinical threshold, single-leg stance measurement is task-intrinsically unstable, and the BlazePose-vs-reference bias is not stable across sessions ( $ICC = 0.094 [-0.22, 0.40]$ ), raising caution against fixed-offset calibration in longitudinal monitoring until larger retest studies confirm bias stability. These clearly characterized failure modes define the boundaries within which single-camera BlazePose can serve as an accessible screening tool for telerehabilitation, while highlighting the need for larger-sample validation, 3D motion-capture benchmarking, and clinical-population studies before routine clinical adoption. Until larger-scale and clinical-population validation is completed, BlazePose-based knee flexion measurement should be deployed in telerehabilitation only as a complement to—not a replacement for—periodic in-person clinical assessment, with explicit awareness that any apparent session-to-session change may reflect measurement variability rather than true clinical change.

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## Conflicts of Interest

The authors declare no conflicts of interest. No commercial pose-estimation or motion-capture vendor was involved in the study design, data collection, analysis, or manuscript preparation.

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## Author Contributions

Xiaotian (Alex) Yue: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Writing – Original Draft, Writing – Review & Editing, Visualization, Project Administration. YiXuan Gu: Investigation (Kinovea annotation, Rater B). Bo Yu: Supervision, Resources.

## Data Availability

The anonymized summary datasets generated and analyzed during the current study are available from the corresponding author upon reasonable request. The BlazePose analysis pipeline used the open-source MediaPipe PoseLandmarker (Heavy model, mediapipe v0.10.33), available from Google's MediaPipe documentation. Kinovea is freely available from the official Kinovea project website. Raw video data cannot be shared due to participant identifiability (faces visible in lateral-view recordings).

## Ethics Approval

This study was approved by the Academic Ethics and Integrity Review Committee of Shanghai Sanda University (approval no. 2026006, April 14, 2026). The study was conducted in accordance with the Declaration of Helsinki. All participants provided informed consent prior to enrollment.

## Abbreviations

2D: two-dimensional  
3D: three-dimensional  
ACL: anterior cruciate ligament  
BMI: body mass index  
CI: confidence interval  
FFprobe: FFmpeg multimedia stream probe  
HDR: high dynamic range  
ICC: intraclass correlation coefficient  
IRB: Institutional Review Board  
LoA: limits of agreement  
MAE: mean absolute error  
OA: osteoarthritis  
RMSE: root mean square error  
ROM: range of motion  
SD: standard deviation  
SEM: standard error of measurement  
SLS: single-leg stance  
STS: sit-to-stand

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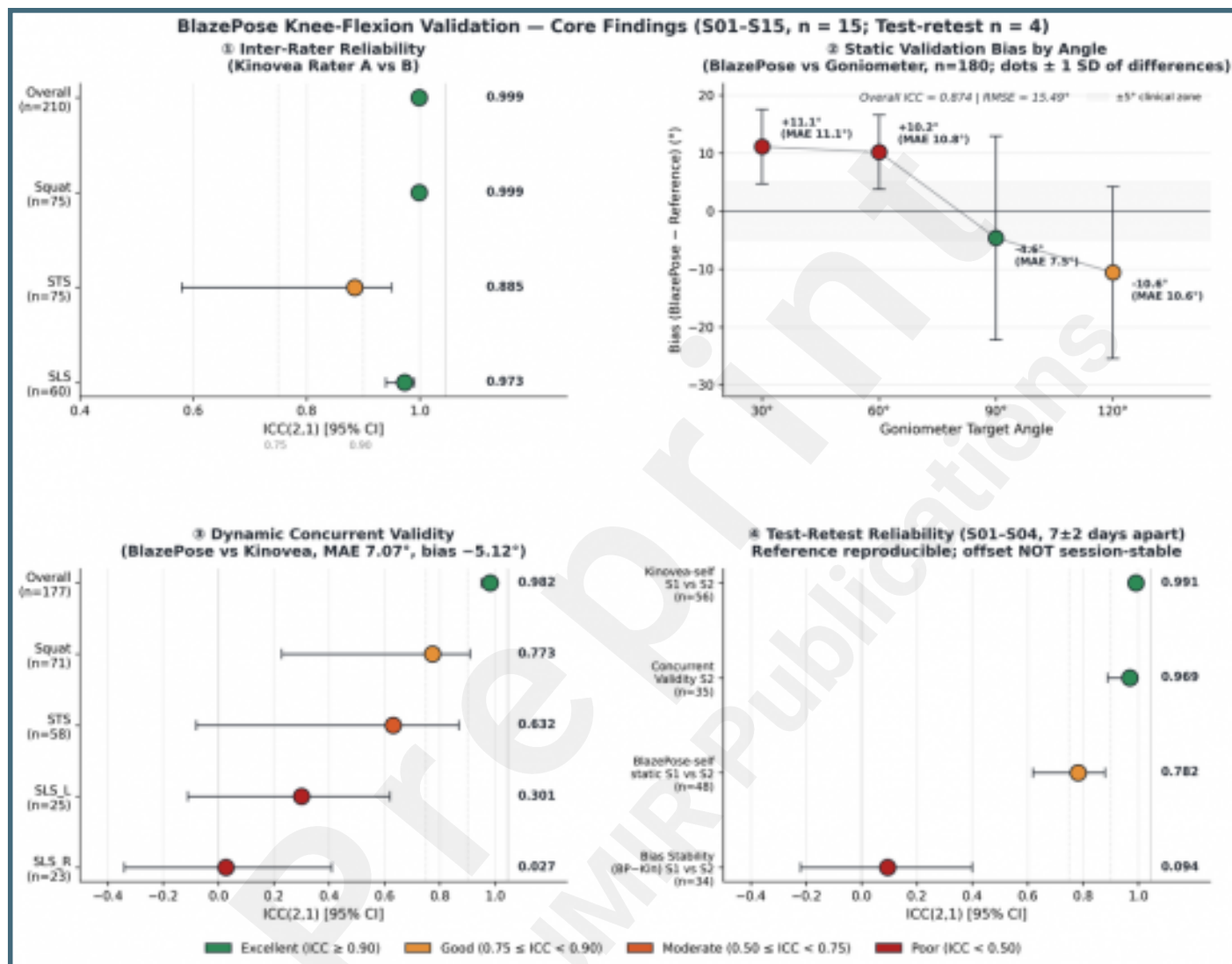
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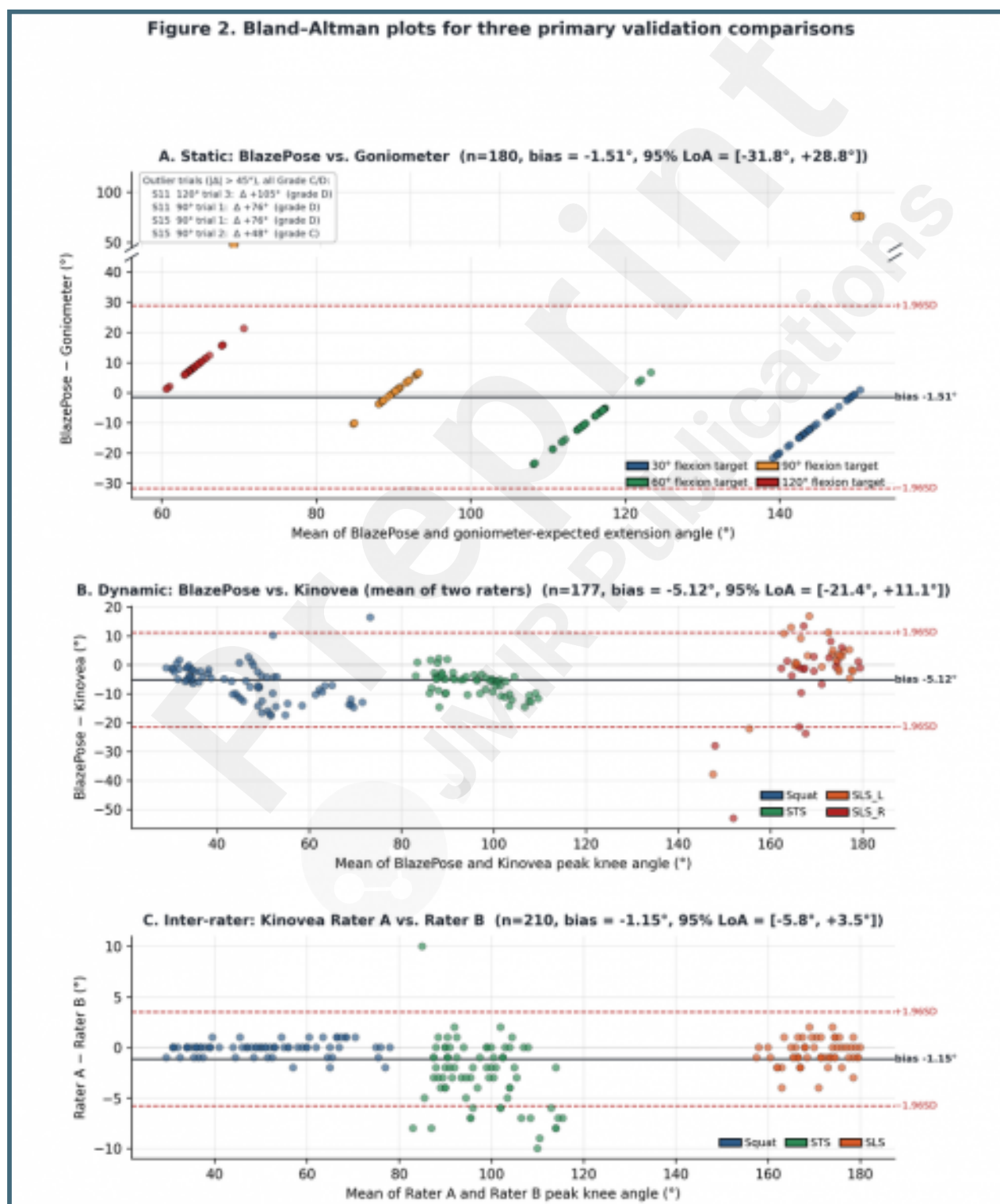
## Supplementary Files

## Figures

Overview of the four core findings. Panel 1 inter-rater reliability of Kinovea (Rater A vs. B) by task. Panel 2 angle-dependent bias of BlazePose static measurements. Panel 3 BlazePose-vs-Kinovea concurrent validity by task. Panel 4 test-retest reliability across four sub-analyses. Color coding: green = excellent (ICC  $\geq 0.90$ ), amber = good (0.75–0.90), orange = moderate (0.50–0.75), red = poor ( $< 0.50$ ). Error bars = 95% CI.



Bland–Altman plots for the three primary validation comparisons. (A) Static BlazePose vs. goniometer ( $n = 180$ ; cross-sectional cohort), colored by target flexion angle. The vertical spread reflects the angle-dependent bias pattern (overestimation at 30°/60° flexion, underestimation at 120°). Panel A uses a broken y-axis: four trials (S11, S15; both Grade C/D) with differences exceeding 45° are shown in a separate zone above the axis break, while the main band of 176 trials is displayed at full resolution below. (B) Dynamic BlazePose vs. Kinovea ( $n = 177$  after confidence  $> 0.3$  filter), colored by task; systematic ~5° underestimation. (C) Inter-rater Kinovea Rater A vs. Rater B ( $n = 210$ ); tight agreement overall, with task-specific spread highest in STS. Solid line = bias; dashed lines = 95% limits of agreement.



## **Multimedia Appendixes**

1. Supplementary Tables S1–S3, including per-subject BlazePose detection confidence distributions, test–retest per-subject breakdowns, and subject-level sensitivity analysis for concurrent validity ICCs.

URL: <http://asset.jmir.pub/assets/e3fa1205eda14a2c4a45977f0bcbb9e4.docx>

